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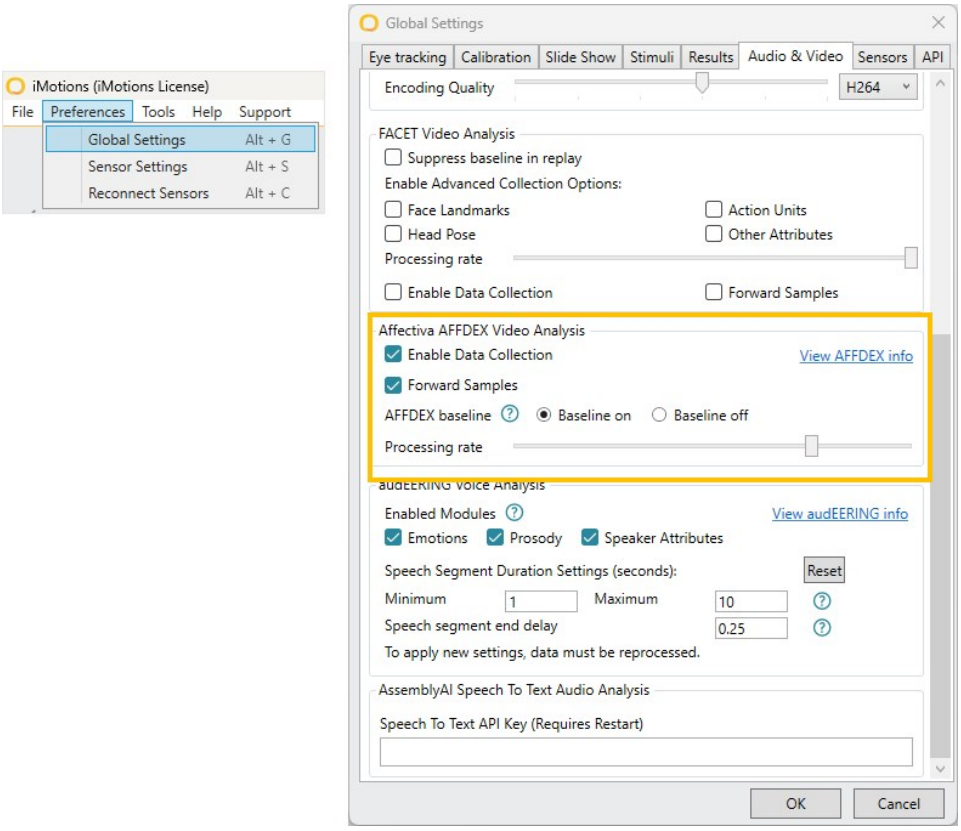
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Enabling Affectiva AFFDEX in iMotions

How to enable Facial Expression Analysis (FEA) with Affectiva AFFDEX.

Enabling AFFDEX

1. Enable your Respondent Camera in iMotions Global Settings (Audio & Video tab)
2. Under the *Affectiva AFFDEX* section in the bottom check the *Enable Data Collection* button.
3. Click 'OK' to save settings.



Enabling AFFDEX in iMotions Global Settings (Audio & Video tab)

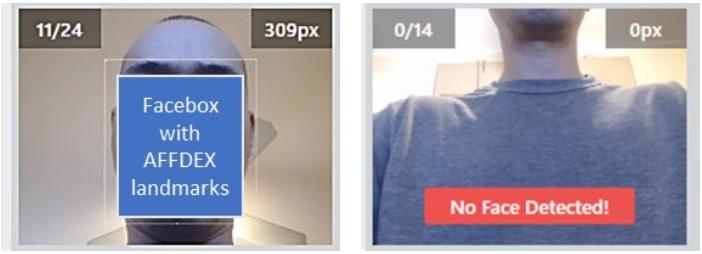
Viewing AFFDEX processing

AFFDEX Overlay in Respondent Camera widget

Once AFFDEX is enabled, the AFFDEX overlay will appear in the Respondent Camera widget:

- When a face is detected, a Facebox and AFFDEX facial landmarks are overlaid on the respondent camera video
  - 'No Face Detected!' warning indicates that AFFDEX can not detect a face, due to respondent position or camera focus.
- Top-left: AFFDEX processing rate/camera capture rate
  - Processing rate shows how many Frames Per Second are currently being processed in real-time, over the Respondent Camera capture frame rate. Typical processing for a standard 30 Hz camera, is 10-20 AFFDEX frames per second. These values vary with camera and computer performance.
- Top-right: Facebox width in camera video pixels
  - Facebox width indicates the width of the white box that appears around the detected face (0px if no face is detected). The Facebox width should be at least 100px.

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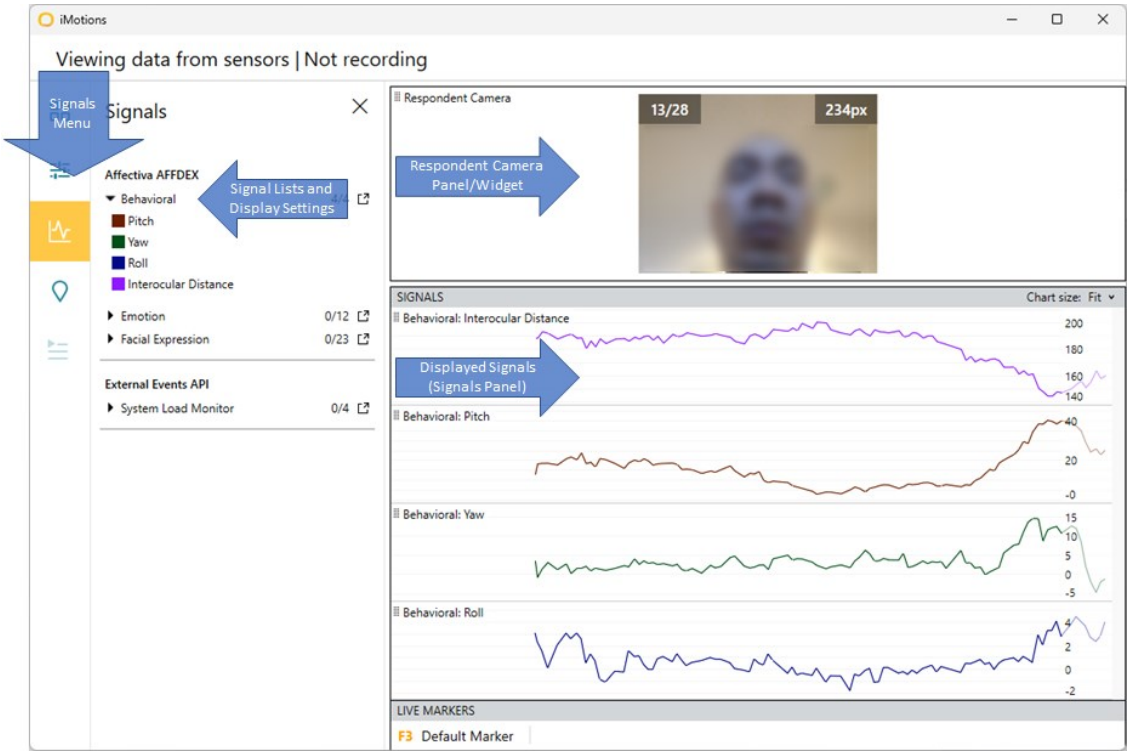


AFFDEX overlay in Respondent Camera widget

AFFDEX Signal in Sensor Preview window

Once AFFDEX is enabled, AFFDEX signals will appear in the Sensor Preview window in the Signals Menu (accessed on the left of the window).

1. Click on 'Sensor Preview'
2. Open the Signals Menu, where 'Affectiva AFFDEX' should be available.
3. Click to the left of any signal to display in the Signals Panel (along the preview timeline)
4. Check that the graphs are streaming live and that the data fluctuates as expected.



Sensor Preview window with Affectiva AFFDEX signal

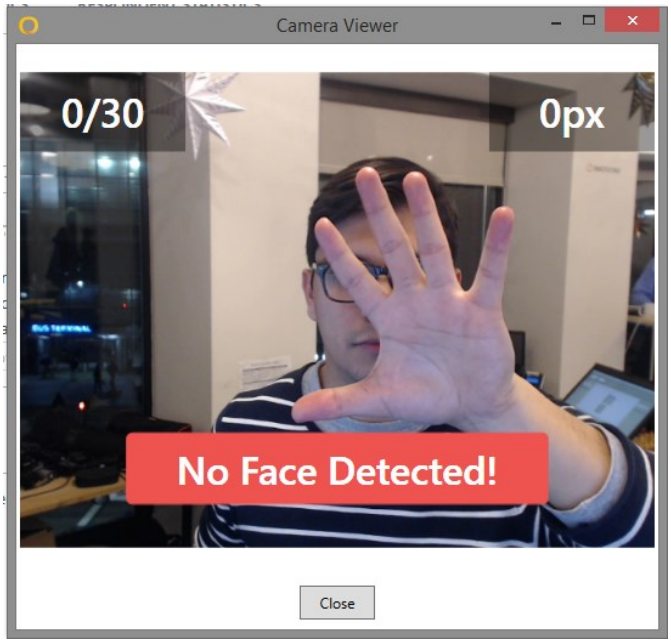
Best Practices for AFFDEX processing

AFFDEX processing relies upon video recording quality, as it relates to the face you want to analyze. Several factors contribute to better recording quality for AFFDEX.

- Lighting and Background Environment are critical factors for AFFDEX processing. Best results are obtained with uniform, diffused illumination of the face against a dark, contrasting background.
    - Bright background lightning, detected by the respondent camera, will 'wash out' the video image - where the camera adjust to the bright light and loses sensitivity to light from the face. More specifically, the face detection is lose if the average pixel brightness for the face falls below 30 on a scale from 0 (black) to 255 (white).
    - Objects in the background, around the face, can interfere with AFFDEX processing. Avoid capturing other faces and/or salient objects with the respondent camera.
- Camera-Respondent Position/Focus is also critical for AFFDEX processing. The Respondent Camera should be set at eye level and directed to the front of the respondent's face. Movement or rotation of the face, away from this position, may result in loss of face detection.
    - It is very likely (if not necessary) that respondents will move a little or a lot - this should be accommodated by the camera position.

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    - The Facebox should be at least 100px, but not take up the entire video frame – this will lower the performance of face detection algorithms.
- Head Movement/Pose is ideally between 5-10 degrees up/down and 40 degrees left/right for AFFDEX 5.1 or newer (25 degrees for older AFFDEX versions). Head tilt rotations are generally accommodated with face processing.
- Respondent Compliance can facilitate AFFDEX processing.
    - Respondent instructions, study task demands, and comfort should be considered, to ensure they can/will remain within view of the Respondent Camera.
    - Respondents should ensure that their hair, hands, clothing, hats, sunglasses, or other objects do not block the camera view. Talking, chewing gum, drinking/eating, phone use, and other distractions should be avoided.
- Camera Quality also affect data quality. Most cameras allow at least 640x480 video resolution at 20-30 frames per second. Higher resolution and/or frame rates may be beneficial, but may also create processing/memory demands to consider.



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